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Normative compliance: **Codes and standards applicable at the project site, to be stated by the responsible engineer**

⚠ Investment, maintenance and electricity costs were not informed: the economic result shows only the savings on the water bill; payback and NPV are not meaningful until costs are entered.

⚠ The grid emission factor was not informed: the emissions from pumping electricity are not meaningful until it is entered.

EXECUTIVE TECHNICAL REPORT (A4 LANDSCAPE)

Triton — Rainwater Harvesting | Project: **Example - tutorial house in New York (ERA5 1991-2020) [costs to be entered by the user]**

CLIMATE SITE	CLIMATE DATASET	SERIES PERIOD	ECONOMIC HORIZON
New York / USA	Observed series	01/01/1991 – 31/12/2020	20 years

RECOMMENDED VOLUME	POTABLE-WATER SAVINGS	NPV (20 YEARS)	ESTIMATED PAYBACK
14,000 L	**28.0%**	**Not calculated**	**Not calculated**
Lower Cistern	Mains-Water Replacement	Net Present Return	Not calculated (discounted)

1. BUILDING & HYDROLOGY

Technical Parameter	Value
Rainfall Series	**NewYork.csv**
Catchment Area	**150.0 m²**
Runoff Coeff. (C)	0.85
First-Flush Discard	2.0 mm
Population Served	4 people
Target Rainwater Demand (28%)	**168 L/day**

2. SIMULATED WATER BALANCE

Operational Indicator	Performance
Cistern Volume	**14,000 L**
Potable-Water Savings	**28.0%**
Mean Rainwater Use	**167.9 L/day**
Mean Potable-Water Use	432.1 L/day
Overflow Volume	137.8 L/day
Full Supply (100%)	**99.9% of days**
No Supply (0%)	**0.1% of days**

3. ECONOMIC FEASIBILITY

Financial Indicator	Result
CAPEX Investment	**Not calculated**
Gross Savings, Year 1	**$ 300.05 / yr**
Simple Payback	Not calculated
Discounted Payback	Not calculated
NPV (20 Years)	**Not calculated**
IRR (% p.a.)	**Not calculated**

📌 Technical Opinion on Water Engineering & Sustainability

⚠ Technical Responsibility & Validation Notice

Triton is a computational sizing-support tool. **The licensed professional is responsible for** validating the executive assumptions, network interferences and compliance with local legislation.

Under the recorded assumptions, the rainwater system shows modelled potable-water savings of **28.0%**; the economic feasibility was not calculated in this study. Sizing follows the technical standards and regulations applicable at the project site.

Rainfall: weather data from Open-Meteo.com (CC BY 4.0), based on the ERA5 reanalysis of the Copernicus Climate Change Service (Hersbach et al., 2020).

Engineer in Charge / Professional
Professional registration / Licence number